CECS 361 Spring 2026

Stage 2: First Progress Report

Custom 16-bit RISC SoC for Digital Lock with VGA Status Display

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Abstract

This project proposes the design and implementation of a custom 16-bit RISC system on chip on the Nexys A7 100-T Board. The system will include a simple processor core, instruction ROM, data RAM and memory-mapped peripherals for switches, push buttons, LEDs, seven segment displays, and VGA output. The processor will execute a digital lock application in which a user enters a code through board inputs, and the system responds by displaying lock status messages such as locked, access granted, access denied, and lockout. The purpose of this project is to demonstrate processor design, memory interfacing, peripheral control, and hardware/software interaction in a visible and interactive embedded environment. The completed design will be verified through simulation of individual modules and demonstrated on the Nexys A7 board with real time visual output.

Introduction

This project implements a custom 16-bit RISC processor on the Nexys A7-100T FPGA board to control a digital lock system with VGA status output. The system demonstrates processor design, memory-mapped input/output, embedded control, and hardware verification. The processor executes a program stored in ROM to read user inputs, compare a passcode, update lock state, and drive peripheral outputs including LEDs, seven segment display, and VGA.

Background and preliminaries

Scope and Concept

The following scope of this project includes:

1. Custom 16-bit CPU basics
2. Memory mapped input and output
3. Nexys A7 100T board peripherals
4. VGA basics
5. Verification strategy

Some of the concepts for this project include

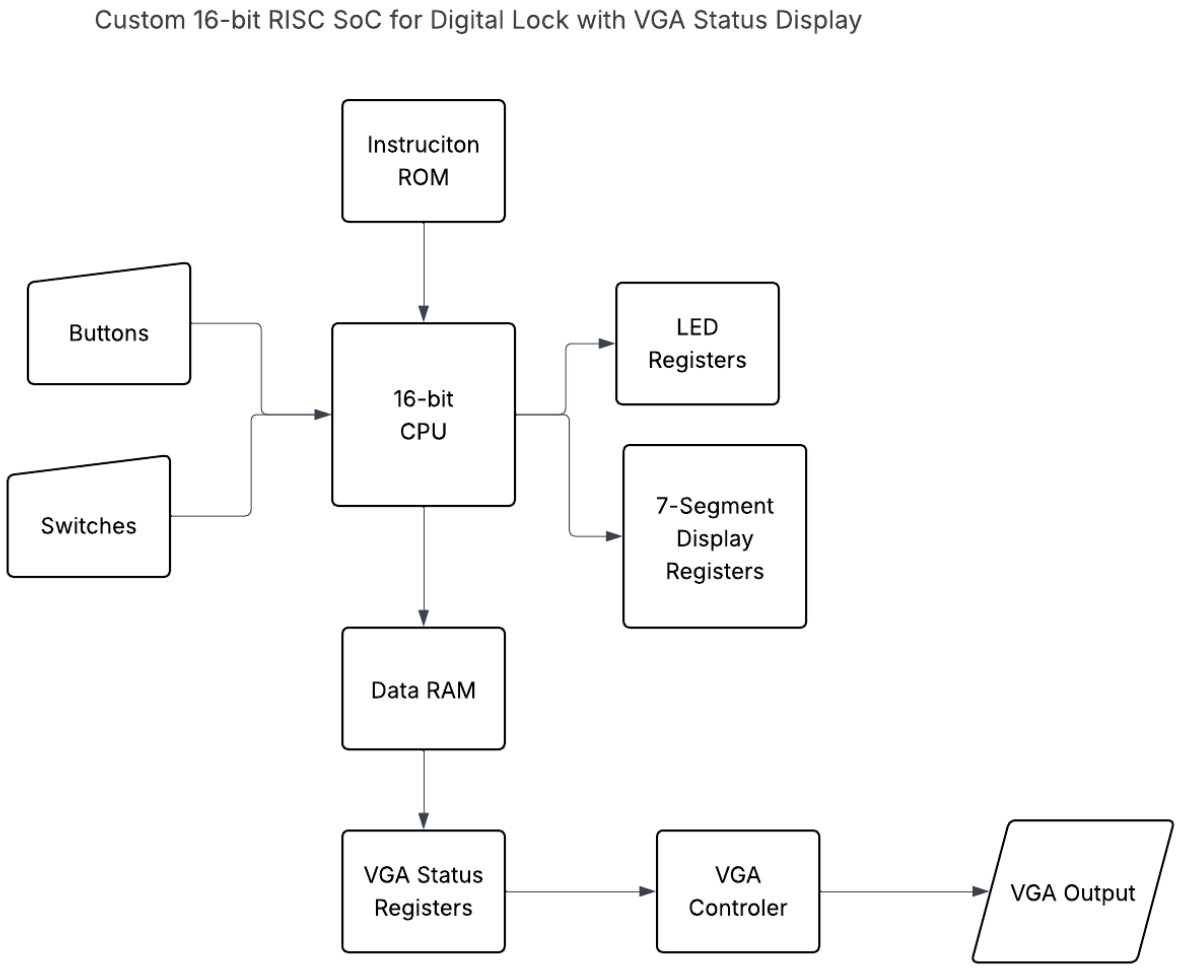
1. Custom CPU Architecture: PC, ALU, registers, control
2. Memory-mapped peripherals: CPU accesses devices through addresses
3. VGA fundamentals: VGA module generates sync/display timing, and CPU only controls the status codes
4. Verification: Testbench simulation for CPU, memory, and peripherals

Research and Resources

The following listed are resources and inspirations used

1. Nexys A7 Board documentation
2. Verilog/Vivado tutorials for ROM, RAM, and block design
3. VGA timing references
4. CPU arhictecture references
5. Course notes on HDL design and verification
6. FPGA Prototyping by Verilog Examples
7. Fundamentals of Computer Architecture and Design
8. Hardware Design Verification: Simulation and Formal Method Based Approaches

Design Diagram



What obstacles are you encountering

The main obstacles have to do with starting off the design. Where to even start, and through trial and error of keeping everything simple. Whether if this has to do with having a simple CPU design, or handling I/O peripherals.

Implementation

The project is currently in the architecture and early implementation stage. The CPU data path, memory map, and peripheral interface have been defined. Initial Verilog modules for instruction ROM, data RAM, and selected I/O registers have been initialized. The lock-control program flow has been drafted, and the VGA interface approach has been simplified so that the CPU writes status codes to a dedicated register instead of directly generating video timing. The CPU is still in its infancy. It will compile, but it is not yet a real processor.

Percentage and Desecription

Member	Description	Percentage
Justin Natividad	Testing, documentation, VGA, CPU	40%
Gabriel Dao	CPU data path, control unit, VGA	30%
Mikko Taitague	Input/Output, LED, switches, seven segment	10%
Matthew Oredenain	VGA, ROM, RAM,	20%